

Chromospheric Ionization and Evaporation

A 1.5D field-aligned two-fluid model of the partially-ionized chromosphere

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Outline

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- 2 Photoionization
- 3 Rates in the C7 Atmosphere
- 4 Cooling
- 5 Chromospheric Evaporation

The chromosphere bridges two regimes

Partially ionized gas: $f \equiv \rho_i / \rho_{\text{tot}}$ runs from $\lesssim 10^{-4}$ at the temperature minimum to ~ 1 at the top of the transition region.

Dense lower chromosphere

collision-dominated, near **LTE / Saha**

- ionization set by Saha equilibrium
- requires collisional ionization balanced by its exact inverse, **three-body recombination**
- $\frac{n_e n_i}{n_n} = \Phi(T_e)$

Rarefied middle / upper chromosphere

out of equilibrium

- ioniz./recomb. time $>$ dynamical time
- f set by *history*, not Saha
- radiation-driven balance: photoionization $\leftrightarrow$ radiative recombination

[Carlsson & Stein 2002; Leenaarts et al. 2007]

Ionization channels

$$\Gamma_{\text{ion}} = \underbrace{n_e n_n (S_i + S_{\text{CR}})}_{\text{collisional}} + \underbrace{n_n P_{\text{phot}}}_{\text{photo}}$$

- Direct ground-state (Voronov):** $S_i(T_e) = 2.91 \times 10^{-14} \frac{U^{0.39} e^{-U}}{0.232 + U} \text{ m}^3 \text{ s}^{-1}$,
 $U = \chi_H / k_B T_e$.
 At $T \sim 5\text{--}6 \text{ kK}$, $U \sim 26\text{--}31$, so $S_i \propto e^{-U}$ is exponentially tiny — **negligible until the transition region**. [Voronov 1997]
- Multilevel / Rydberg-ladder** $S_{\text{CR}} = \alpha_c \Phi / n_e$: stepwise excitation up the level ladder ($\chi_n = \chi_H / n^2 \rightarrow 0$, no Boltzmann penalty). $\sim 10^4\text{--}10^5 \times$ ground state; the ionizer that reaches Saha at the base. [Bates, Kingston & McWhirter 1962; Johnson 1972]
- Photoionization** ($\text{Ly}\alpha + \text{Balmer-continuum}$): the **dominant chromospheric ionizer**; energy comes from radiation $\Rightarrow$ no χ_H drain on the gas. [Carlsson & Stein 2002]

Recombination channels

$$\Gamma_{\text{rec}} = \underbrace{n_e^2 \alpha_r}_{\text{radiative}} + \underbrace{n_e^2 \alpha_c}_{\text{three-body (collisional)}}$$

- **Radiative** $\text{H}^+ + e \rightarrow \text{H} + \gamma$, Hummer case-B fit:
 $\alpha_r = 2.7 \times 10^{-19} (T_e/10^4 \text{ K})^{-0.75} \text{ m}^3 \text{ s}^{-1}$. Binding energy escapes as an (optically-thin) Ly-continuum photon. [Hummer 1994]
- **Three-body / collisional-radiative** $\text{H}^+ + e + e \rightarrow \text{H}(n) + e$, rate $\propto n_e^2 n_i$. Capture into high- n Rydberg levels then cascade $\Rightarrow \sim 5 \times 10^4 \times$ the ground-state estimate; grows with n_e .
 $\alpha_c \approx 5.6 \times 10^{-27} (k_B T_e [\text{eV}])^{-9/2} n_e \text{ cm}^3 \text{ s}^{-1}$. [Hinnov & Hirschberg 1962; Stevefelt et al. 1975]
- **Neutral / molecular** ($\propto n_e n_i n_n, \propto n_i n_n^2$): comparable to radiative *only* at the dense base; carried as quantified uncertainty. [Drawin 1968; Cretu et al. 2022]

Equilibrium: one formulation, two limits

Setting $df/dt = 0$ (dropping P_{phot} for clarity):

$$(1 - f) S_i = f [\alpha_r + \alpha_{3b} f n_{\text{tot}}]$$

Low density ($\alpha_c \ll \alpha_r$)

$$\frac{f}{1 - f} = \frac{S_i}{\alpha_r}$$

density-independent kinetic balance.

High density ($\alpha_c \gg \alpha_r$), with $\alpha_{3b} = S_i/\Phi$

$$\boxed{\frac{f^2}{1 - f} = \frac{\Phi(T_e)}{n_{\text{tot}}}}$$

the [Saha equation](#) — pure thermodynamics.

Detailed balance holds *level by level* ($\alpha_{3b}(n) = S_i(n)/\Phi_n$), so the collisional ionization / three-body pair has Saha as its *exact* equilibrium. A single-level model with the $S_{\text{CR}} = \alpha_c \Phi / n_e$ closure recovers the LTE limit *by construction*.

Photoionization — frozen field, inverted from C7

$$\text{H} + \gamma \rightarrow \text{H}^+ + e, \quad \Gamma_{\text{ion}}^{\text{phot}} = n_n P_{\text{phot}}$$

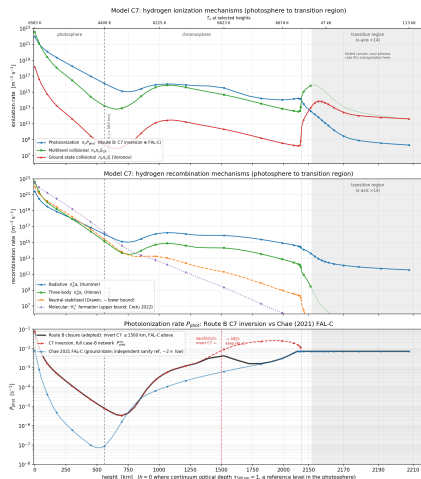
- We do *not* solve radiative transfer. The chromospheric photoionizing field is nearly time-independent — its radiation temperature varies only $\sim 1\text{--}2\%$ even through shocks (Balmer-continuum photons come from the slowly-varying deep layers). [Carlsson & Stein 2002]
- $\Rightarrow$ freeze the radiative rate into a function of height alone (the “fixed radiative-rate” recipe). [Sollum 1999; Leenaarts et al. 2007]

C7-context inversion. Solve the local balance of the *full* network on C7’s tabulated (n_p, n_e, n_n, T) so that C7 is an equilibrium fixed point:

$$P_{\text{phot}} = \frac{n_p}{n_n} (\alpha_r + \alpha_c) n_e - (S_i + S_{\text{CR}}) n_e, \quad n_p = n_{\text{H}} - n_{\text{H1}} \quad [\text{Avrett \& Loeser 2008, Tab. 26}]$$

- Blend above $z \approx 1500$ km to the frozen FAL-C $R_{ik}(z)$ of [Chae 2021].
- Profile: $R_{ik} \sim 7 \times 10^{-8} \text{ s}^{-1}$ at T_{min} , rising to $\sim 7 \times 10^{-3} \text{ s}^{-1}$ at the chromosphere top.

Dominant mechanisms across the C7 column



Layer	Ionization	Recombination
Lower (Saha)	multilevel S_{CR} (Rydberg ladder)	three-body α_c ($\alpha_c/\alpha_r \approx 3$)
Mid/Upper (NEQ)	photoionization (Ly α +Balmer)	radiative α_r ($\alpha_c/\alpha_r \lesssim 0.04$)
TR	direct Voronov S_i	radiative (weak)

Organizing principle: n_e sets the *rate* competition (3-body $\propto n_e$ vs. radiative), T_e sets the *equilibrium* (Saha) and the e^{-U} suppression of direct ionization.

In evaporation, both edges activate:

- **Voronov** S_i switches on at high T (upflow $\rightarrow$ coronal T).
- **Three-body** α_c dominates in dense gas (condensation downflow).

Energy sinks: ionization potential + radiation

1. Ionization-potential bookkeeping ($\chi_H = 13.6 \text{ eV}$).

- Collisional ionization *drains* the electron pool: $-\Gamma_{\text{ion}}^{\text{col}}\chi_H$ (photoionization does *not* — its energy came from radiation).
- Three-body recombination is **superelastic**: $+\Gamma_{\text{rec}}^{\text{col}}\chi_H$ handed to the third electron, stays as heat.
- Net non-radiative budget $= -\Gamma_{\text{ion}}^{\text{col}}\chi_H + \Gamma_{\text{rec}}^{\text{col}}\chi_H \rightarrow 0$ in LTE (detailed balance) — *no spurious sink* in the deep chromosphere.
- Radiative recombination removes one χ_H Ly-continuum photon (optically-thin case-B loss).

Cooling can be switched off (no-cooling) to isolate ionization energetics; it is **on by default**.

Radiative cooling: optically thick + optically thin

2a. Optically-thick chromospheric line cooling [Carlsson & Leenaarts 2012]

H I Ly α , Ca II H & K + IR triplet, Mg II h & k — formed out of LTE, escape probability $\ll 1$:

$$Q_{\text{rad}}^X = L_X(T_e) E_X(\tau_X) \frac{N_{X^*}}{N_X}(T_e) A_X \frac{N_H}{\rho} n_e \rho, \quad X \in \{\text{H I, Ca II, Mg II}\}$$

$E_X(\tau_X)$ = empirical escape probability vs. Ly α optical depth / column mass (downward integration); effective for $T \lesssim 3 \times 10^4$ K.

2b. Optically-thin TR / coronal loss [Klimchuk et al. 2008]

$$Q_{\text{thin}} = n_e n_H \Lambda(T), \quad \Lambda_{\text{peak}} \approx 4 \times 10^{-35} \text{ W m}^3 \text{ at } \log T \approx 5.1$$

Switched on smoothly above $\sim 2 \times 10^4$ K (no double-count). Lets the lower TR re-radiate the imposed coronal conductive flux $q(T)$.

Both advanced on the [implicit side](#) of the operator split (Stage R) — $\tau_{\text{rad}} \sim 1\text{--}10$ s falls below the explicit CFL step.

Explosive evaporation — the physics to reproduce

[Fisher, Canfield & McClymont 1985]

- **Energy-flux threshold:** explosive when heating exceeds what the upper chromosphere can radiate near the 10^5 K loss peak: $F_{\text{crit}} \approx 7 \times 10^9 \text{ erg cm}^{-2} \text{ s}^{-1} = 7 \times 10^6 \text{ W m}^{-2}$. Above $\rightarrow$ **explosive**; below $\rightarrow$ *gentle*, conduction-driven.
- **Timescale:** $3kT/Q \lesssim L_0/c_s$ — heat faster than hydrodynamic expansion (~ 10 s).
- **Velocity ceiling:** upflow $v \leq 2.35 c_s$ of the evaporated (coronal- T) plasma.
- **Driver:** thick-target nonthermal electron beam, deposited where it stops in the upper chromosphere; matching downward condensation recoil.

Requirement: the radiative sink **must be active** — the threshold *is* the heating-vs-radiation competition.

Setup: boundary, initial, and deposited-energy conditions

Initial condition

- Model C7 chromosphere column [Avrett & Loeser 2008]
- discrete-HSE base; let it settle one acoustic time

Boundary conditions

- lower: physical $V = 0$ + numerical diffusion
- upper: **opened outer boundary** (Neumann V , free supersonic outflow) — the evaporated plasma must leave the top
- ambient coronal conductive flux $q(T)$, RTV (negligible vs. beam)

Deposited electron energy

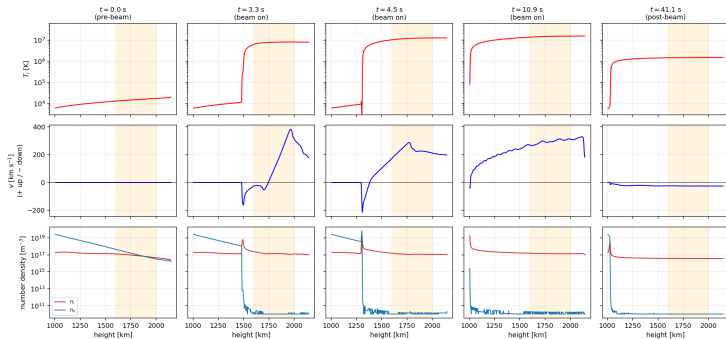
- volumetric beam-heating source (not a hot wall)
- thick-target stopping is sub-grid $\Rightarrow$ deposit over $\Delta h_{\text{beam}} \approx 80$ km, weighted $\propto n_{\text{tot}}$, normalized to column flux F_e
- into the electron/ion thermal pool; applied just *before* cooling
- $t_{\text{on}} = 2$ s, $\tau = 10$ s, 1 s cosine ramps

Two runs

- **Explosive:** $F_e = 5 \times 10^7 \text{ W m}^{-2}$
- Gentle control: $F_e = 1 \times 10^6 \text{ W m}^{-2}$

Evolution — results

Explosive chromospheric evaporation (Fisher et al. 1985), refined 500-cell run
beam $F_e = 5 \times 10^{10} \text{ erg cm}^{-2} \text{ s}^{-1}$ (above threshold), on 2–12 s, deposited over 1600–2000 km (shaded)



- Upper chromosphere heated to $\sim 10^6$ – 10^7 K; cannot radiate the deposited flux.
- Explosive upflow of evaporated plasma into the corona (~ 100 s of km s^{-1} , $\leq 2.35 c_s$), with a downward condensation front ($\sim -150 \text{ km s}^{-1}$) below it.
- Gentle control: stays cool, only a weak ($\lesssim 30 \text{ km s}^{-1}$) upflow — the threshold isolated.

Neutral-profile oscillation — a trace-species effect

What is seen. In the evaporating front the gas is driven to $f \rightarrow 1$, so the **neutral becomes the trace species** ($n_n \rightarrow 0$); its profile develops oscillations while n_{tot} stays smooth.

Why. Where a species' density craters faster than its pressure relaxes, $T = p/(2nk_B)$ blows up. This bites at *both* edges:

- evaporation front: $f \rightarrow 1$, $n_n \rightarrow 0$ (trace *neutral*);
- condensation downflow: recombination drives $f \rightarrow 0$, $n_i \rightarrow 0$ (trace *ion*) — the mirror image, a transient T_i spike in the dense lower chromosphere.

Diagnostic: n_{tot} smooth while f and the trace density oscillate $\Rightarrow$ **not** a hydro/limiter artifact, but trace-species overheating fed back through the cell-local Stage-E solve.

Fix. Downward-only cap of the trace pressure to the dominant fluid's temperature (single-fluid limit) with a $2\times$ deadband — kills the oscillation, preserves the physical -184 km s^{-1} condensation flow, never touches V .

Key references I

- Avrett & Loeser 2008, *ApJS* 175, 229 — Model C7 reference atmosphere.
- Fisher, Canfield & McClymont 1985, *ApJ* 289, 425 — explosive chromospheric evaporation.
- Carlsson & Stein 2002, *ApJ* 572, 626 — NEQ ionization; frozen photoionizing field.
- Carlsson & Leenaarts 2012, *A&A* 539, A39 — optically-thick chromospheric cooling recipe.
- Hummer 1994, *MNRAS* 268, 109 — radiative recombination (case-B).
- Voronov 1997, *ADNDT* 65, 1 — collisional ionization.
- Hinnov & Hirschberg 1962, *Phys. Rev.* 125, 795; Stevefelt et al. 1975, *Phys. Rev. A* 12, 1246 — three-body / CR recombination.
- Bates, Kingston & McWhirter 1962, *Proc. R. Soc. A* 267, 297; Johnson 1972, *ApJ* 174, 227 — multilevel collisional-radiative rates.
- Chae 2021, *JASS* 38, 83 — frozen-field photoionization rate $R_{ik}(z)$.
- Leenaarts et al. 2007, *A&A* 473, 625 — NEQ hydrogen ionization.
- Klimchuk et al. 2008 — optically-thin coronal loss function.
- Drawin 1968, *Z. Phys.* 211, 404; Cretu et al. 2022, *Phys. Rev. A* — neutral/molecular channels.